

Before you start reading: general assumptions and recommendations

Assumptions

When reading through the following project specs, please make the following assumptions:

- Unless otherwise specified as optional, all attributes are required
- Unless otherwise specified, if given a list of potential values, choices should be limited to that list
 - If a set of values is listed with "and/or", combinations of those values are possible, while just "or" indicates only a single value is possible

Implementation recommendations

Please be mindful of the following when planning and designing your implementation:

- You should create normalized schemas, and minimize the use of NULL attributes whenever and wherever possible
- Ensure that you store non-numeric data that appear as numbers (such as street numbers, phone numbers, postal codes, etc.) as strings and not numeric data types
- Avoid "catch-all" forms with unnecessary inputs that the user would leave empty or NULL
- You do not need to be concerned about handling concurrent operations that could conflict and introduce inconsistencies in your database
- All data manipulations and calculations should occur within database queries wherever it is possible
 - The number of queries to produce any required output (such as a report) should be kept to a minimum
- You may implement the project UI as a traditional standalone application (e.g., Java GUIs or Python's TkInter) or as a web application (e.g., using web scripting languages like PHP or JSP).
 - Creating an API and performing raw operations via API calls, and as manual interactions, will not be considered sufficient for a "user interface"
 - The UI does not need to be overly complex or stylish, it should "just work"
 - Your project will not be graded on its aesthetic appeal, but rather on its ability to meet the FUNCTIONAL specification requirements presented here and on the correctness of all results and expected output
- Do not create any additional functionality that is not mentioned in this specification (such as email notifications, etc.) or attempt to enhance your final product beyond what the specification requires
 - Adding unwanted functionality can divert attention from what the specification is asking you to deliver, and this diversion will impact your grade!

(the project specification begins on the next page)

Midtown Motors

A Georgia Tech alumnus, Yel O. Jacket, has decided to go into the business of selling used cars and started a company named Midtown Motors. For now he would just like a simple application to update inventory when purchasing and selling, keep track of financial and inventory information, and allow customers visiting the showroom to search the inventory for vehicles with desired properties. At this time, there aren't enough staff to take pictures of the cars, so only the attributes of the cars will be stored in the database. Along with the functions to support operations, he has identified some reports so he can track how well the new business is doing.

Functionality and Users

Midtown Motors will have a public-facing interface (similar to the landing page of a company website), accessible without any login and implements a basic interface to search for vehicles.

The front page will also have login functionality available and, once logged-in, the company staff members can access additional features not available to public users. These additional features will be described in further detail later, but it is important to remember to distinguish between public users and logged-in users and what they may/may not access. Furthermore, these are the following categories of users that will login, with various permissions:

- Acquisition specialists, who buy vehicles and add them to inventory, as well as enter parts orders; they also record information about the seller, car's previous owner.
- Sales agents, who will only have access to searching available inventory, will be recording information about sales transactions and buyers.
- Operating managers, who can view inventory, purchase history, sales transactions, parts order history, and reports.
- And the Owner of Midtown Motors, who has access to everything and can perform any activity in the system (a combination of all permissions).

The comprehensive details of responsibilities and functions of all categories of logged-in (aka privileged) users will be provided below in the corresponding sections of this document.

Since this is a prototype system, it will not be necessary to have an interface for adding/registering users and granting them appropriate permissions. The database administrator will manually add users and set permissions as needed using SQL commands. All logged-in users will be identified using a unique username determined by the database administrator (DBA) and will login with their username and a password assigned by the DBA; the DBA will also set the first and last name of the user (it is acceptable that passwords in our system will be stored as plaintext).

Throughout this specification, purchase always refers to Midtown Motors acquiring a vehicle. Sale or sell always refers to Midtown Motors transferring a vehicle to a buyer.

Operational Details

There are a variety of people and things involved in the day-to-day operations of Midtown Motors. You should build a database schema that facilitates storing the information needed for these processes.

Vehicles

Vehicles are tracked on a variety of characteristics. First, each vehicle has a unique alphanumeric Vehicle Identification Number (VIN). Next, the type of vehicle and vehicle condition is stored. The list of vehicle types is in the appendix and should be updatable by the database administrator in case new types of vehicles are invented; the condition should be chosen from these fixed choices: Excellent, Very Good, Good, Fair or Rough. The manufacturer name is also stored, and a list of valid manufacturer names is provided in the appendix of this specification. This list of manufacturers may change, so you should ensure the list can be updated within the database by the DBA. The model name and vehicle model year must also be stored, and these will be free-form entered by the user, with the restriction that model year cannot exceed the current year plus one. (A model year's vehicles are usually available midway through the previous calendar year. For example, someone might sell to Midtown Motors a 2020 model year vehicle in 2019, but it is impossible to sell a 2021 model year car in 2019 since that year's models do not exist yet.) The year entered must include century digits ("1999" is acceptable, but "15" is not). The fuel type of the vehicle is also tracked, which can be one of the following: Gas, Diesel, Natural Gas, Hybrid, Plugin Hybrid, Battery, or Fuel Cell. Of course, the color of the car is also an important detail, and a list of generic color names that can be chosen for a vehicle is also provided in the appendix. A car may have multiple colors, for example, silver and red. The list of colors is not expected to change. Horsepower (a whole number) is also stored for each vehicle, as well as the drivetrain layout, which can only be set to 4WD, AWD, FWD or RWD. Finally, an optional notes field can be used to enter additional information such as what accessories or equipment the car has or anything else that might be interesting for a customer searching for a vehicle.

Customers

Sellers and buyers (combined, referred to as "customers", because they can both buy and sell) can be either an individual person or a business. For all customers, their address (street address/city/state/postal code) and their phone number are collected. Customers also have the option of providing an email address so that Midtown Motors can stay in touch with them electronically. If a customer is an individual, their first and last names, along with their social security number (SSN, which is unique), will be recorded. If the customer is a business, the business' tax identification number (similar to a Social Security number, and also unique) and business name, along with the first and last name of a primary contact and their title (such as owner, fleet manager, etc.), are recorded.

Sellers

Vehicles are purchased by Midtown Motors from sellers, and the vehicle should link back to the seller from whom it was purchased. It is safe to assume that Midtown Motors will purchase and sell a single vehicle only once, but each seller may sell multiple vehicles. Purchase prices are entered for each purchase manually by the acquisition specialist handling the transaction; an acquisition specialist will also determine the condition of the vehicle and assign one of the

previously listed condition values. The purchase date should be tracked so that the company has a mechanism determine how long the vehicle remained in the inventory.

A customer may act as both a seller and buyer at different times; if it already exists, the same customer record should be used for both activities.

Parts

Some of the purchased vehicles need new parts before they can be sold. An acquisition specialist will determine what parts are necessary, order parts from vendors, and record each parts order in the system. Each parts order can include one or several kinds of parts, but must be associated with one vendor, so you will need to keep track of the name of the vendor (we can assume that vendor's names are unique), their address (street address/city/state/postal code) and phone number, and an internal parts order number will be generated. Midtown Motors' generates parts order numbers by combining the VIN with the ordinal of the order, and padded to three digits – so the first parts order for a vehicle with VIN ABCDEFGHIJ1234567 would be ABCDEFGHIJ1234567-001, the second ABCDEFGHIJ1234567-002, etc.

For each part in a parts order, you will need to track status (ordered or received or installed), a description of the part, vendor's part number (which may be alphanumeric), and the unit price of the part. If multiple quantities of a specific part are needed (such as all four wheels of a vehicle), the quantity will be tracked for that part within the parts order, and all units of that part number will share the same status. The total cost of a parts order is the sum of total costs of each part (including quantity) within the parts order. Part status will be manually updated by an acquisition specialist; however, the status of the part cannot be reverted to a previous status (e.g. "installed" cannot be changed to "ordered"). The status for each kind of part within a parts order must be tracked individually as different kinds of parts on the same order are rarely received all together; however, we can assume that parts of the same kind (such as wheels) are always all ordered/received/installed at the same time. It is also possible for any vehicle to have multiple parts orders from the same vendor. Because part prices may change, you should not worry about maintaining a static list of part numbers and prices. A vehicle cannot be returned within any public search results or be sold if it has any parts pending or not installed (in other words, all parts ordered for a given vehicle must be installed before this vehicle can become accessible through public search).

Buyers

Vehicles are sold to buyers in a sale transaction which is organized and executed by a sales agent. The sale price is calculated as 125% of the original purchase price (the price Midtown Motors paid to purchase the vehicle) combined with 110% of any parts costs also associated with the vehicle. Just as with the purchase, the vehicle should have a link to the customer to whom the vehicle was sold, and it is possible (and good for business) that one buyer can buy several vehicles. Should a buyer acquire several vehicles at the same time, they would still each be handled in a separate per-vehicle sale transactions. The sale date should be tracked to determine when a vehicle leaves inventory.

Application Functionality

Public Access

The only feature accessible to the public is searching for vehicles. Because of this, the initial landing page of our application should offer search functionality to the public as well as an option to login which will be used only by staff members.

This page should also initially display somewhere prominently the total number of vehicles available for sale in the system, that is, vehicles without any pending parts. Searching on the following criteria should be allowed:

- Vehicle type
- Manufacturer
- Year
- Fuel type
- Drivetrain layout
- Color
- Keyword, which searches the manufacturer, model, year and notes fields. Any vehicle record that matches the entered keyword (either entirely or as a substring, case-insensitive) for any of those fields should be returned.

For fields other than Keyword it may be appropriate to use dropdowns to provide choices to the user. You should not allow making multiple selections for these fields, only a single value selection should be allowed in each dropdown. Results must match all search options that are entered. No selection for the dropdown implies that all values for the corresponding field should be acceptable.

If no vehicles meet the search criteria, a message should be displayed: “Sorry, it looks like we don’t have that in stock!”

If there are vehicles that match the search criteria, you should return the following attributes for each vehicle in the search results:

- VIN
- Vehicle type
- Manufacturer
- Model
- Year
- Fuel type
- Drivetrain layout
- Color(s) – be sure if a car has multiple colors, that it only returns a single result row but that all colors are listed
- Horsepower
- Sale Price

These results should be sorted by VIN in ascending order. Users must not be allowed to change or override the sort order. Users can then select an individual result, which will open a detail page that includes the VIN, Vehicle type, Manufacturer, Model, Year, Fuel type, Drivetrain layout, Color(s), Horsepower, Sale price, and the Notes fields for the selected vehicle.

Privileged Access

As noted previously, users who are employees of Midtown Motors will have access to additional features necessary to perform their job duties. Remember that you do not need to provide any interface for creating or registering users and granting them privileges, as this will be done

manually in the database by the DBA. Privileged users will login using their username and password. Ideally, all users will start on the public-facing search screen, which provides a login option, and after logging in, the page will get updated to include access to the appropriate functionality.

One area of common functionality is the ability to look up and add customers to the system. However, this functionality should only be available when performing a purchase or sale transaction and should not be independently accessible. Looking up a customer can be done using either SSN or tax ID. If no result is found, then the option to add a customer is provided, and based on the customer type, the appropriate fields (as described earlier in this specification) should be input into the system.

The search page for all privileged users should provide additional option of searching by VIN. Search by VIN should always be case insensitive.

Acquisition Specialists

After an acquisition specialist logs in, he/she will be given access to an “Add Vehicle” button or link, that will allow them to add a new vehicle, which is being purchased. On the add vehicle form, the specialist will need to search for or add a customer to link the purchase of the vehicle to its seller. After selecting a seller for the car, the new vehicle form will gather all the relevant vehicle details such as VIN, vehicle type, condition, purchase price, etc.; the purchase date should also be recorded. After submitting the data and successfully adding the vehicle to the database, the acquisition specialist will be taken to the detail page for the vehicle. (What the detail page looks like for acquisition specialists will be described further on.)

Acquisition specialists will need to also pull up any of the vehicles currently in the inventory and will do that using the search screen, which should show them somewhere on the search screen the number of vehicles currently with parts pending along with the number of vehicles available for sale. Specialists can search on the same criteria as public users, along with the option to search by VIN. Unlike public search, however, the results for an acquisition specialist should also include any vehicle that has not been sold, even those with parts pending. Selecting a specific vehicle will load that vehicle’s detail page.

The acquisition specialist’s view of the detail page is similar to the detail page shown to public users and should show the same information but should additionally include fields for the original purchase price and the total of all parts costs. A newly added vehicle will show \$0 total for parts because it has no parts ordered yet. There should also be a section for parts, which will list each part that was ordered for the car. This list should include all relevant details for each part: part number, description, vendor, parts order, part unit price, quantity, and status. There should also be a mechanism for updating the status of an individual part from ordered to received, from received to installed, or directly, from ordered to installed. A part’s status cannot be changed to a previous status, and once it is installed, its status can no longer be updated. An “add parts order” link or button should also be provided to access the Add Parts Order form.

If a part needs to be entered, the acquisition specialist will access the parts order form and enter the information for that part (its elements were described previously in the Parts section). Similar to how customers are handled, you should have a mechanism on the parts order form for searching and adding vendors to link them to a parts order. Since different parts may be included

in the same order, you will need to provide a mechanism for entering multiple kinds of parts into a parts order.

Sales Agents

Sales agents will start, after logging in, on the search page, with the same layout as a public search, with the added option to search by VIN, and like public users, with the results only including vehicles with no pending parts on order. Upon loading the detail page for a vehicle, the sales agent will see the same detail page that customers do, with an added button or link to sell the car.

Using this button/link, sales agents can look up a customer (or add them if a customer is not found) and confirm the sale; the date of each sale will be recorded. The sale price of the vehicle cannot be changed – Midtown Motors’ philosophy doesn’t believe in bargaining over vehicle prices and customers will enjoy knowing that the price listed for a car is the exact price they will pay without any added hassle.

Operating Managers

Managers have view-only access to all information along with reports (which will be described in their own section). Like acquisition specialists and sales agents, after logging in, operating managers start on the search screen, which will display somewhere the number of vehicles currently with parts orders pending, along with the number of vehicles available for sale, with the same search options as a public search, and can also search by VIN. They should be provided with additional option for filtering by sold vehicles, unsold vehicles, or all vehicles. When filtering by unsold or all vehicles, all unsold vehicles will be returned regardless of parts status.

When viewing a vehicle detail page, managers will see all information for the vehicle – including all of the seller’s contact information (everything except their SSN or tax ID number), the name (first and last) of the acquisition specialist that purchased the car, the original purchase price, the purchase date, the total cost of parts, and a parts section listing details for all parts just like it would be shown on an acquisition specialist’s view. In addition, if the vehicle has been sold, the buyer’s contact information (everything except their SSN or tax ID number), sale date, and the sales agent’s name (first and last) will be displayed.

Owner

As stated previously, the Owner has access to the complete functionality of the system, must be able to view all information and reports, and should be able to do any activity described previously in this specification. Essentially, the Owner’s login will allow her to do anything an operating manager, acquisition specialist, or sales agent can do, keeping in mind any context for business processes. (For example, the vehicle detail page will show all information like it does for operating managers, and the Owner will also be able to sell a vehicle or add parts to it) .

Reports

Reports will be visible to the Owner and and operating managers. Access to the reports should be via a link, button, or dropdown menu that can be displayed on the initial search page for users that are allowed access to reports.

Seller History

This report will provide information about all sellers from whom Midtown Motors purchased vehicles. It will include the following elements: the name of the seller (first name and last name for individual or business name for business, which should be displayed as a single column, not in two different columns for each seller type), the total number of vehicles, the average purchase price for the vehicles that were purchased from them by Midtown Motors, the average total quantity of all parts per vehicle, and the average overall cost of parts per vehicle; all average values should be calculated over all vehicles purchased from this seller. The report should be sorted by total number of vehicles purchased descending, followed by average purchase price ascending. In addition, any seller who shows an average of five or more parts (in total quantity of all parts column) , or where the average cost of parts is \$500 or more, should have their resulting row highlighted with a red background to indicate that they may be selling lower quality vehicles and that Midtown Motors may want to avoid purchasing from them in the future.

Average Time in Inventory

This report, based on the difference between the vehicle sale and purchase dates, will display, by vehicle type, the average amount of time a vehicle remains in inventory, in days. When counting days spent in the inventory, the first and last day should be calculated as Sale Date – Purchase Date, with one date added. If a vehicle was purchased and sold on the same day, such as October 15, it will be counted as one day in inventory (), not zero, but if purchased on October 15 and sold on October 16, it will count as two days (). If a vehicle type has no sales history, the report should display “N/A” for that vehicle type. The results should be ordered by vehicle type in ascending order.

Price Per Condition

This report will display, by vehicle type, and for each condition (Excellent, Very Good, Good, Fair, Rough), the average price paid for cars that Midtown Motors has purchased. If a vehicle type or condition has never been purchased, the report should display “\$0” for that result. This should be a matrix report (there should not be a row for each combination of vehicle type and condition), with condition as the rows, in descending order (Excellent Rough) and vehicle type as the columns, sorted in alphabetical order.

Parts Statistics

This report will be used to negotiate better prices with parts vendors. In this report, you should list: the vendor’s name, the total quantity of all parts supplied by that vendor, and the total dollar amount spent on parts coming from this vendor. The report should be sorted by vendor name in ascending order.

Monthly Sales

This report will be the most frequently used report and will have two parts. First, a summary page, which lists for all sale transactions, by year and month, the total number of vehicles sold, the gross sales income, and the total net income (which is sale price less the vehicle purchase price and less total expenses on parts bought for this vehicle). If a year or month does not have sales, it can be excluded from this report. The results will be ordered by year and month descending, with the most recent year and month as the first result.

For each year/month result, a drilldown report for that year and month must be accessible. Based on the sales data for that year and month, the drilldown will display the top performing sales agents by showing the sales agent's first and last name, the number of vehicles they sold in that year and month and their total sales for that year and month. To determine who is the top sales agent for the month, the drilldown will be sorted by total vehicle numbers descending followed by total sales descending. (In other words, in the event of a tie where two or more sales agents have sold the same number of vehicles, the sales agent who has sold the highest dollar value will be considered the top sales agent.)

Appendix

Manufacturers

Acura	FIAT	Lamborghini	Nio
Alfa Romeo	Ford	Land Rover	Porsche
Aston Martin	Geely	Lexus	Ram
Audi	Genesis	Lincoln	Rivian
Bentley	GMC	Lotus	Rolls-Royce
BMW	Honda	Maserati	smart
Buick	Hyundai	MAZDA	Subaru
Cadillac	INFINITI	McLaren	Tesla
Chevrolet	Jaguar	Mercedes-Benz	Toyota
Chrysler	Jeep	MINI	Volkswagen
Dodge	Karma	Mitsubishi	Volvo
Ferrari	Kia	Nissan	XPeng

Colors

Aluminum	Beige	Black	Blue	Brown	Bronze	Claret
Copper	Cream	Gold	Gray	Green	Maroon	Metallic
Navy	Orange	Pink	Purple	Red	Rose	Rust
Silver	Tan	Turquoise	White	Yellow		

Vehicle Types

Convertible	Coupe	Minivan	Other	Sedan	SUV	Truck	Van
-------------	-------	---------	-------	-------	-----	-------	-----

Change History

Version	Date	Description
1.0	1/26/2026	Initial version of specification